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| **Completed** | Monday, 5 October 2026, 5:17 PM |
| **Duration** | 7 mins 45 secs |
| **Grade** | **14.00** out of 15.00 (**93.33**%) |

Question **1**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By sorting data in a single pass
- b. By breaking the input into smaller parts and merging them
- c. By comparing elements sequentially
- d. By using the bubble-up method

The correct answer is: By breaking the input into smaller parts and merging them

Question **2**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

- a. Quick sort
- b. Bubble sort
- c. Insertion sort
- d. Heap sort

The correct answer is:
Insertion sort

Question **3**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,70,80,90,120,150,200,300,400,470
- b. 80,150,200,470,40,90,300,400,70,120
- c. 40,80,90,150,200,300,400,470,70,120
- d. 200,470,80,150,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **4**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. order()
- b. arrange()
- c. sort()
- d. sorted()

The correct answer is: sort()

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

- a. Bubble Sort
- b. Insertion Sort
- c. Merge Sort
- d. Quick Sort

The correct answer is: Bubble Sort

Question **6**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. order()
- b. arrange()
- c. sorted()
- d. sort()

The correct answer is: sorted()

Question **7**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Sorting the entire list sequentially
- b. Dividing the input into smaller subproblems
- c. Comparing each element with the others
- d. Combining sorted sublists

The correct answer is: Dividing the input into smaller subproblems

Question **8**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It only works with integer arrays
- b. It requires external libraries
- c. It sorts data out of the box efficiently
- d. It is less efficient than custom sorting algorithms

The correct answer is: It sorts data out of the box efficiently

Question **9**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Searching
- b. Duplicates
- c. Selection
- d. All of the above

The correct answer is: All of the above

Question **10**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Binary Search
- b. Bubble Sort
- c. Linear Search
- d. Merge Sort

The correct answer is: Merge Sort

Question **11**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Being the most efficient sorting algorithm
- b. Bubbling up the largest element to its correct position with each pass
- c. Using the divide-and-conquer approach
- d. Sorting data in a non-sequential manner

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **12**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
b. A sorting algorithm is stable if it preserves the order of non-duplicate keys
c. A sorting algorithm is stable if it preserves the order of all keys
d. A sorting algorithm is stable if it preserves the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **13**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Arranging data in a particular format
b. Searching for data in a list
c. Inserting data into a list
d. Deleting data from a list

The correct answer is: Arranging data in a particular format

Question **14**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Divide and conquer
b. Dynamic programming
c. Backtracking
d. Greedy method

The correct answer is:
Divide and conquer

Question **15**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It builds a sorted array one element at a time
b. It compares adjacent elements and swaps them if necessary
c. It repeatedly finds the minimum element and moves it to the sorted part of the list
d. It divides the list into two halves, sorts each half, and then merges them

The correct answer is: It divides the list into two halves, sorts each half, and then merges them